Retrieval-Augmented Generation (RAG)

Learning notes for document-loading and RAG practice

What is RAG?

Retrieval-Augmented Generation, or RAG, is a technique that gives a language model relevant information from an external knowledge source before it generates an answer. The model uses the retrieved information as context.

Why use RAG?

A language model may not know private documents or recently updated information. RAG can connect the model to an external knowledge base without retraining the model every time the documents change.

Main Stages

The main stages are ingestion, retrieval, augmentation, and generation. During ingestion, documents are loaded, parsed, split into chunks, converted into embeddings, and stored. During retrieval, relevant chunks are searched for a user's question. During augmentation, the retrieved text is added to the prompt. During generation, the language model writes an answer.

Documents and Metadata

A document commonly contains page_content and metadata. Page content stores the actual text. Metadata stores details such as source filename, page number, author, or creation date. Metadata can help filter search results.

Embeddings and Vector Stores

An embedding is a numeric representation of text. Texts with similar meanings often have vectors that are close in the embedding space. A vector store saves embeddings and supports similarity search.

Limitations

RAG depends on the quality of the source documents and retrieval process. If the wrong chunks are retrieved, the answer may be incomplete or incorrect. Retrieved context should be checked and the final answer should cite sources when possible.